Iniziato	martedì, 7 gennaio 2020, 15:07
Stato	Completato
Terminato	martedì, 7 gennaio 2020, 15:36
Tempo impiegato	28 min. 43 secondi
Punteggio	10,50/15,00
Valutazione	**21,00** su un massimo di 30,00 (**70**%)

Domanda **1**
Risposta corretta
Punteggio ottenuto 1,00 su 1,00

In order to reduce the dimensionality of a dataset, which is the advantage of Multi Dimensional Scaling (MDS), with respect to Principal Component Analysis (PCA)

Scegli un'alternativa:

a. MDS can be used also with categorical data, provided that the matrix of the distance is available, while PCA is limited to vector spaces

b. MDS can be used with categorical data after a transformation in a vector space

c. MDS requires less computational power

d. MDS can work on any kind of data, while PCA is limited to categorical data

Domanda **2**
Parzialmente corretta
Punteggio ottenuto 0,50 su 1,00

Which is the purpose of discretisation?

Scegli un'alternativa:

a. Increase the number of distinct values in an attribute, in order to put in evidence possible patterns and regularities

b. Reduce the range of values of a numeric attribute, to make all the attributes more comparable

c. Reduce the number of distinct values in an attribute, in order to put in evidence possible patterns and regularities

d. Reduce the number of distinct values in an attribute, in order to increase the efficiency of the computation The increased efficiency of the computations can be a byproduct, but the main reason is to put in evidence possible patterns and regularities

Domanda **3**

Risposta errata

Punteggio ottenuto 0,00 su 1,00

Given the two binary vectors below, which is their similarity according to the Simple Matching Coefficient?

a b c d e f g h i j

1 0 0 0 1 0 1 1 0 1

1 0 1 1 1 0 1 0 1 0

Scegli un'alternativa:

a. 0.1

b. 0.5

c. 0.2

d. 0.3

Domanda **4**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the following statements regarding the discovery of association rules is true? (One or more)

Scegli una o più alternative:

a. The support of a rule can be computed given the confidence of the rule

b. The support of an itemset is anti-monotonic with respect to the composition of the itemset

c. The confidence of an itemset is anti-monotonic with respect to the composition of the itemset

d. The confidence of a rule can be computed starting from the supports of itemsets

Domanda **5**

Risposta errata

Punteggio ottenuto 0,00 su 1,00

Given the definitions below:

- TP = True Positives
- TN = True Negatives
- FP = False Positives
- FN = False Negatives

which of the formulas below computes the *precision* of a binary classifier?

Scegli un'alternativa:

a. (TP + TN) / (TP + FP + TN + FN)

b. TN / (TN + FP)

c. TP / (TP + FP)

NO d. TP / (TP + FN)

Domanda **6**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

A Decision Tree is...

Scegli un'alternativa:

a. A tree-structured plan of tests on multiple attributes to forecast the target

b. A tree-structured plan of tests on single attributes to obtain the maximum purity of a node

c. A tree-structured plan of tests on single attributes to forecast the cluster

d. A tree-structured plan of tests on single attributes to forecast the target

Domanda **7**

Risposta errata

Punteggio ottenuto 0,00 su 1,00

Given the two binary vectors below, which is their similarity according to the Jaccard Coefficient?

a b c d e f g h i j

1 0 0 0 1 0 1 1 0 1

1 0 1 1 1 0 1 0 1 0

Scegli un'alternativa:

a. 0.375

b. 0.5

c. 0.1

d. 0.2

Domanda **8**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which is different from the others?

Scegli un'alternativa:

a. K-means

b. Decision Tree

c. Dbscan

d. Expectation Maximisation

Domanda **9**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

After fitting DBSCAN with the default parameter values the results are: 0 clusters, 100% of noise points. Which will be your next trial?

Scegli una o più alternative:

a. Reduce the minimum number of objects in the neighborhood

b. Decrease the radius of the neighborhood

c. Reduce the minimum number of objects in the neighborhood and the radius of the neighborhood

d. Increase the radius of the neighborhood

Domanda **10**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

How does *pruning* work when generating frequent itemsets?

Scegli un'alternativa:

a. If an itemset is frequent, then none of its supersets can be frequent, therefore the frequencies of the supersets are not evaluated

b. If an itemset is not frequent, then none of its subsets can be frequent, therefore the frequencies of the subsets are not evaluated

c. If an itemset is frequent, then none of its subsets can be frequent, therefore the frequencies of the subsets are not evaluated

d. If an itemset is not frequent, then none of its supersets can be frequent, therefore the frequencies of the supersets are not evaluated

SUPER
V/
ITME
V/
SUB

Domanda **11**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

What measure is maximised by the Expectation Masimisation algirithm for clustering?

Scegli un'alternativa:

a. The support of a class

b. The *likelihood* of a class label, given the values of the attributes of the example

c. The likelihood of an example

d. The likelihood of an attribute, given the class label

Domanda **12**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the following preprocessing activities is useful to build a Naive Bayes classifier if the independence hypothesis is violated

Scegli un'alternativa:

a. Feature selection

b. Standardisation

c. Normalisation

d. Discretisation

Domanda **13**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

What does K-means try to minimise?

Scegli un'alternativa:

a. The *separation*, that is the sum of the squared distances of each point with respect to its centroid

b. The *separation*, that is the sum of the squared distances of each cluster centroid with respect tho the global centroid of the dataset

c. The *distortion*, that is the sum of the squared distances of each point with respect to the points of the other clusters

d. The *distortion*, that is the sum of the squared distances of each point with respect to its centroid

Domanda **14**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

What is the *cross validation*

Scegli un'alternativa:

a. A technique to improve the speed of a classifier

b. A technique to obtain a good estimation of the performance of a classifier when it will be used with data different from the training set

c. A technique to improve the quality of a classifier

d. A technique to obtain a good estimation of the performance of a classifier with the training set

Domanda **15**

Risposta errata

Punteggio ottenuto 0,00 su 1,00

The *information gain* is used to

Scegli un'alternativa:

a. select the attribute which maximises, for a given test set, the ability to predict the class value

b. select the attribute which maximises, for a given training set, the ability to predict all the other attribute values

c. select the attribute which maximises, for a given training set, the ability to predict the class value

d. select the class with maximum probability